

Step 4: Automatic Pet Feeder: Word Code Implementation

Step 1: Initialize System

Set up clock, servo motor, food availability sensor, and bowl weight sensor.

Comment: Ensures the device knows the time, can dispense food, can detect if the hopper has food, and can weigh the bowl.

Step 2: Check Feeding Time / Manual Request

- `current_time = get_current_time()`
- If `current_time == 8:00AM` or `current_time == 6:00PM` or `manual_feed_pressed == True`:
- Proceed to Step 3
- Else: wait (loop back to Step 2)

Comment: Triggers feeding at scheduled times or on manual request; otherwise the system idles.

Step 3: Check Food Availability

- If `food_available == True`:
- Proceed to Step 4
- Else: send "Refill Food" alert and loop back to Step 2

Comment: Prevents running the servo when the hopper is empty and notifies the user.

Step 4: Dispense Food & Record Baseline Weight

- `rotate_servo(3s)`
- `bowl_weight_before = get_bowl_weight()`

Comment: Servo runs for ~3 seconds to release a preset portion & immediately record the bowl weight as the baseline for comparison.

Step 5: Wait and Check Bowl Weight

- Wait 10 minutes
- `bowl_weight_after = get_bowl_weight()`
- `bowl_weight_change = bowl_weight_before - bowl_weight_after`

Comment: A decrease indicates food has been eaten.

Step 6: Evaluate Result & Notify

- If `bowl_weight_change >= 5g`:
- Feeding successful
- Else: send "Uneaten Food" alert

Comment: 5g is the threshold from the flowchart & below this, the pet likely didn't eat.

Step 7: Return to Waiting State

- Loop back to Step 2 to monitor the next feeding time or manual request.

Comment: Closes the cycle and keeps the system running continuously